Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport
la
Lucrarea de laborator Nr. 7

Disciplina: Tehnici de programare

Tema: ” Git, pas cu pas .Unealta stricata, reparata cu git”

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Lucrarea A: Git,pas cu pas

Configurarea unui cont GitHub, pe calculator și două depozite publice de exercițiu, exercitiu-git și exercitiu-2, cu mai multe commit-uri trimise pe GitHub. În cuvinte simple: git ține istoria proiectului pe calculatorul tău, iar GitHub ține o copie a ei pe internet.

Lucrarea B: Unealtă stricată, reparată cu git

Aducerea fisierului inventar.py la starea în care rulează și dă exact ieșirea așteptată. Nu o rescrii: o repari câte o greșeală pe rând, citind fiecare eroare, și faci câte un commit după FIECARE reparație. La final, git log --oneline este jurnalul depanării tale.

Scopul Lucrării

Învățarea utilizării programelor Git și GitHub pentru salvarea și organizarea proiectelor, precum și exersarea corectării pas cu pas a erorilor dintr-un cod Python, salvând fiecare modificare făcută.

Mersul Lucrării

Lucrarea A: Git,pas cu pas.

```
PS C:\Users\midon> cd ~
PS C:\Users\midon> git clone https://github.com/Azot-cloud/exercitiu-git.git
Cloning into 'exercitiu-git'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\midon> cd exercitiu-git
PS C:\Users\midon\exercitiu-git> ^C
PS C:\Users\midon\exercitiu-git> cd C:\Users\midon\Uni Projects\TP\Labs
Set-Location : A positional parameter cannot be found that accepts argument 'Projects\TP\Labs'.
At line:1 char:1
+ cd C:\Users\midon\Uni Projects\TP\Labs
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidArgument: (:) [Set-Location], ParameterBindingException
    + FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Users\midon\exercitiu-git> cd "^C
PS C:\Users\midon\exercitiu-git> cd "C:\Users\midon\Uni Projects\TP\Labs"
PS C:\Users\midon\Uni Projects\TP\Labs> cd .\exercitiu-git
```

Figura 1.1 / Git clone (A4)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> python salut.py
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> python salut.py
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> python salut.py
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> python salut.py
Salut, GitHub!
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        salut.py

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git add salut.py
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   salut.py

PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git commit -m "Adaug salut.py"
[main b27aa2b] Adaug salut.py
 1 file changed, 1 insertion(+)
 create mode 100644 salut.py
```

Figura 1.2 / Prima fotografie (A5)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 306.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Azot-cloud/exercitiu-git.git
   77c19b0..b27aa2b  main -> main
```

Figura 1.3 / Git push (A6)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 949 bytes | 49.00 KiB/s, done.
From https://github.com/Azot-cloud/exercitiu-git
   58ad658..c2d285f  main       -> origin/main
Updating 58ad658..c2d285f
Fast-forward
 README.md | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Figura 1.4 Git pull (A7)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> type README.md
# exercitiu-git
Exercitii de git pentru Tehnici de programare

Aici exersez git.
```

Figura 1.5 Înserare în READ.ME (A7)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> python -m venv .venv
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status --short
?? .venv/
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status --short
?? .gitignore.txt
?? .venv/
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status --short
?? .gitignore
?? .venv/
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git status --short
?? .gitignore
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git add .gitignore
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git commit -m "Ignor mediul virtual"
[main ca22f66] Ignor mediul virtual
 1 file changed, 2 insertions(+)
 create mode 100644 .gitignore
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 341 bytes | 341.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Azot-cloud/exercitiu-git.git
   c2d285f..ca22f66  main -> main
```

Figura 1.6 .gitignore (A8)

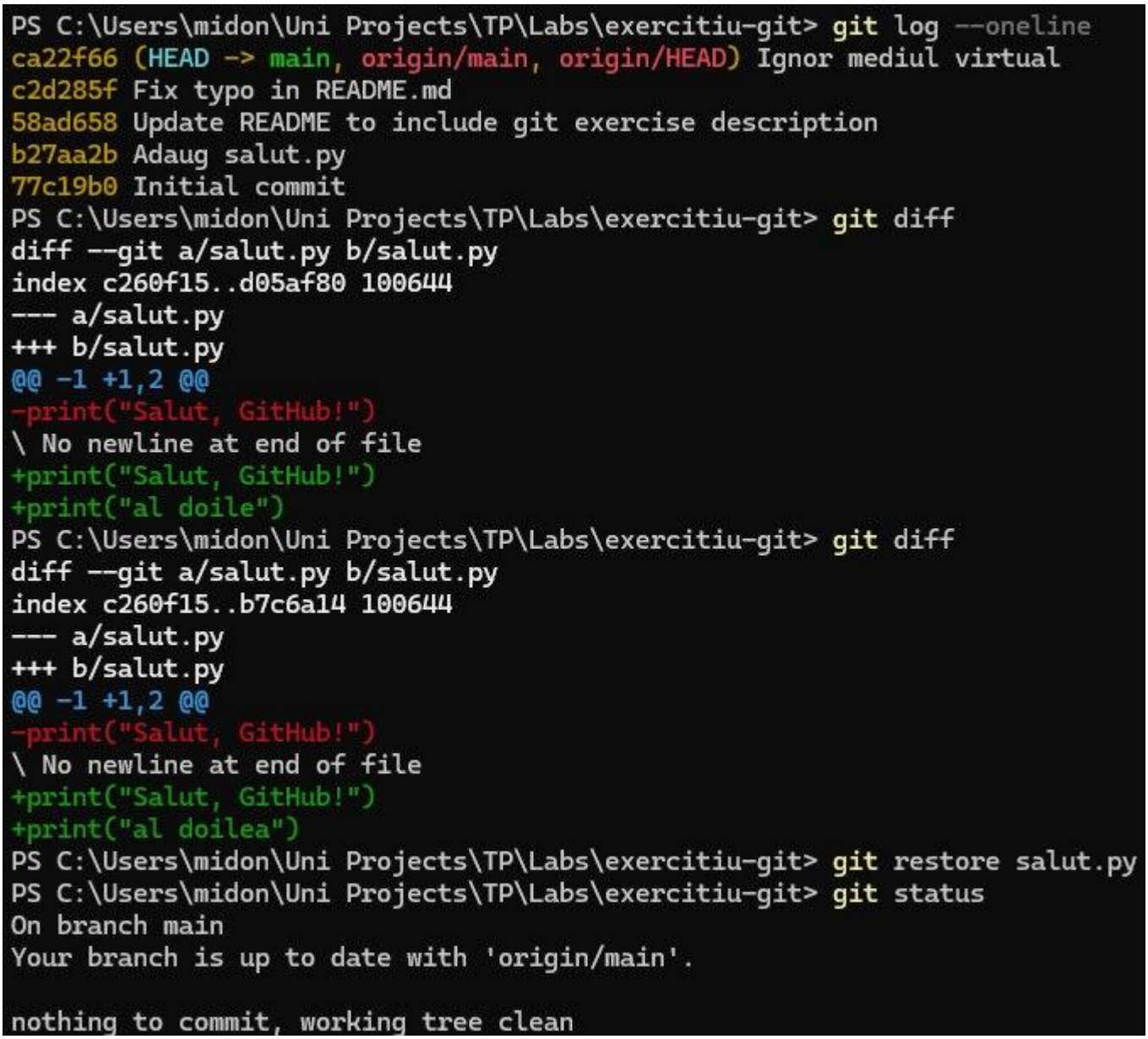


Figura 1.7 log si diff (A9)

```
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-git> cd ~
PS C:\Users\midon> cd "C:\Users\midon\Uni Projects\TP\Labs"
PS C:\Users\midon\Uni Projects\TP\Labs> mkdir exercitiu-2


    Directory: C:\Users\midon\Uni Projects\TP\Labs


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         11/1/2026   6:23 PM                exercitiu-2


PS C:\Users\midon\Uni Projects\TP\Labs> cd exercitiu-2
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git init
Initialized empty Git repository in C:/Users/midon/Uni Projects/TP/Labs/exercitiu-2/.git/
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> code README.md
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git add README.md
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git commit -m "Primul commit"
[main (root-commit) 3c44910] Primul commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git remote add origin https://github.com/Azot-cloud/exercitiu-2.git
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git branch -M main
PS C:\Users\midon\Uni Projects\TP\Labs\exercitiu-2> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 235 bytes | 235.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Azot-cloud/exercitiu-2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Figura 1.8 Al doilea depozit (A10)

Lucrarea B: Unealtă stricată, reparată cu git.

```
PS C:\Users\midon> cd "C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git"
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git> git pull
Already up to date.
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git> mkdir depanare


    Directory: C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         11/1/2026  11:32 PM                depanare


PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git> git add depanare
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git> git commit -m "Pornire: inventar.py stricat"
[main 328431e] Pornire: inventar.py stricat
 2 files changed, 75 insertions(+)
 create mode 100644 depanare/inventar.py
 create mode 100644 depanare/tinte.json
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git> cd depanare
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
  File "C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare\inventar.py", line 8
    def incarca(cale)
                    ^
SyntaxError: expected ':'
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
  File "C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare\inventar.py", line 19
    return None
    ^^^^^^
IndentationError: expected an indented block after 'except' statement on line 18
```

Figura 2.1 Pregătire, în depozitul exercitiu-git (B1)

```
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git pull
Already up to date.
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "SyntaxError: am adaugat :"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "IndentationError: am indentat return None"
[main b8f1caa] IndentationError: am indentat return None
 1 file changed, 2 insertions(+), 2 deletions(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "NameError tinta: numele corect e tinte"
[main 82d8d50] NameError tinta: numele corect e tinte
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "TypeError concatinare str: am adugat str(len())"
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "TypeError concatinare str: am adugat str(len())"
[main 2b70735] TypeError concatinare str: am adugat str(len())
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "SyntaxError lipseste ): am inchis paranteza"
[main 3684bd1] SyntaxError lipseste ): am inchis paranteza
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "KeyError serviciu: am adaugat .get si am transformat in tuple"
[main a6add48] KeyError serviciu: am adaugat .get si am transformat in tuple
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "AttributeError NoneType: am adaugat return la functia pe_servicii(tinte)"
[main f029e2e] AttributeError NoneType: am adaugat return la functia pe_servicii(tinte)
 1 file changed, 1 insertion(+)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "TypeError la prag: type=int in argparse"
[main 697a1ac] TypeError la prag: type=int in argparse
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
ssh: [22]
Peste 1000: []
Raport salvat in raport_inventar.json
```

Figura 2.2 Metodă la fiecare eroare (B2)

```
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "KeyError ('gazda', 'necunoscut') : am schimbat [] cu ()"
[main 5266a3d] KeyError ('gazda', 'necunoscut') : am schimbat [] cu ()
 1 file changed, 7 insertions(+), 6 deletions(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "TypeError not callable"
[main 787d07f] TypeError not callable
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -am "ValueError la prag: am folosit port invalid in peste_prag"
[main b92f630] ValueError la prag: am folosit port invalid in peste_prag
 1 file changed, 1 insertion(+), 1 deletion(-)                                  python inventar.py tinte.json --prag 1000^C\Labs\Lucrarea_de_Laborator_7 (TP)
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> python inventar.py tinte.json --prag 1000
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
ssh: [22]
http: [80, 8080]
mysql: [3306]
necunoscut: [5432]
Peste 1000: ['192.168.56.102', '192.168.56.103', '192.168.56.104']
Raport salvat in raport_inventar.json
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> python inventar.py tinte.json
Tinte citite: 6
sar peste 192.168.56.103 : port invalid abc
ssh: [22]
http: [80, 8080]
mysql: [3306]
necunoscut: [5432]
Peste 1024: ['192.168.56.102', '192.168.56.103', '192.168.56.104']
Raport salvat in raport_inventar.json
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git add raport_inventar.json
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git commit -m "Raportul unei unelte reparate"
[main 47d74a6] Raportul unei unelte reparate
 1 file changed, 22 insertions(+)
 create mode 100644 depanare/raport_inventar.json
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git push
Enumerating objects: 50, done.
Counting objects: 100% (50/50), done.
Delta compression using up to 12 threads
Compressing objects: 100% (49/49), done.
Writing objects: 100% (49/49), 5.21 KiB | 761.00 KiB/s, done.
Total 49 (delta 21), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (21/21), done.
To https://github.com/Azot-cloud/exercitiu-git.git
   ca22f66..47d74a6  main -> main
```

Figura 2.3 (B3 / B4 / B5)

```
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare> git log --oneline
47d74a6 (HEAD -> main, origin/main, origin/HEAD) Raportul unei unelte reparate
b92f630 ValueError la prag: am folosit port invalid in peste_prag
787d07f TypeError not callable
5266a3d KeyError ('gazda', 'necunoscut') : am schimbat [] cu ()
697a1ac TypeError la prag: type=int in argparse
f029e2e AttributeError NoneType: am adaugat return la functia pe_servicii(tinte)
a6add48 KeyError serviciu: am adaugat .get si am transformat in tuple
3684bd1 SyntaxError lipseste ): am inchis paranteza
2b70735 TypeError concatinare str: am adugat str(len())
82d8d50 NameError tinta: numele corect e tinte
b8f1caa IndentationError: am indentat return None
328431e Pornire: inventar.py stricat
ca22f66 Ignor mediul virtual
c2d285f Fix typo in README.md
58ad658 Update README to include git exercise description
b27aa2b Adaug salut.py
77c19b0 Initial commit
PS C:\Users\midon\Uni Projects\TP\Labs\Lucrarea_de_Laborator_7 (TP)\exercitiu-git\depanare>
```

Figura 2.4 Verificare și final push(B6)

CONCLUZII

În urma realizării acestui laborator, am învățat să configurez și să utilizez sistemul Git în combinație cu platforma GitHub pentru o gestionarea mai eficientă a depozitelor de cod și de sincronizarea proiectelor.Astfel, am identificat si remediat eruorile in mod sistematic, înțelegând importanța salvării fiecărei modificări printr-un commit individual.

BIBLIOGRAFIE

1.